

Solutions to Quantum Mechanics: Concepts and
Applications
by Nouredine Zettili

Kieron Eisma

August 2026

*To Dr. Miller Mendoza Jiménez from ETH Zürich
Thank you for inspiring me to do theoretical physics.*

Preface

This is an ongoing project of mine to solve a significant percentage of all the problems in the third edition of Zettili's *Quantum Mechanics* [Zet22]. I frequently refer to [Axl24] as linear algebra is embedded in quantum theory. As always, should there be any errors, please email me:

kiwi.eisma@gmail.com

The only way to understand physics is by *doing* physics. However, working your way through problems is useless without the ability to verify your solutions. There is good reason why books on the subject (usually) do not make the answers public, for immediately looking at the answers the moment you get stuck on a problem is extremely detrimental to one's comprehension of a topic. That said, it is also detrimental to an individual's understanding if they cannot verify their workings. Therefore, it is imperative that you, the reader, use this manual *wisely* and do not give up on a difficult problem. Of course, if you really can't solve something, do consult the solutions, but always remember that the lesson of an exercise is not the exercise itself, but rather the way it forces you to organise your thoughts before solving it.

I hope this manuscript becomes useful to the theoretical physics community (undergraduate students) as I continue to update it with more and more solutions.

Kieron Eisma

ACKNOWLEDGMENTS AND AI DISCLOSURE

ChatGPT 5.5 has been used to typeset this document (converting my working into TeX code), and to generate plots of functions to aid readers' visualisation. All solutions, however, have been completely fabricated by myself.

Contents

1	Origins of Quantum Physics	7
2	Mathematical Tools of Quantum Mechanics	9
	References	27

Chapter 1

Origins of Quantum Physics

Exercise 1.37.

Consider the function

$$\psi(x) = \begin{cases} 1 - |x|, & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

To find the Fourier Transform $\phi(p)$ of this function, consult Exercise 2 on page 161 of [SS03].

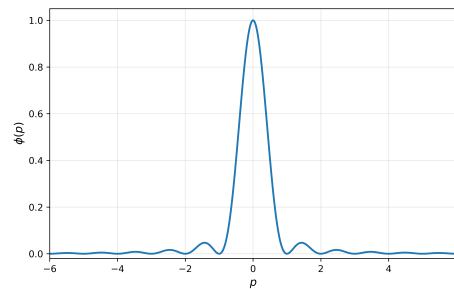


Figure 1.1: *Plot of the Fourier transform $\phi(p)$ of ψ . Plot generated by ChatGPT*

Chapter 2

Mathematical Tools of Quantum Mechanics

Exercise 2.1(a)

For the first two norms, we have:

$$\begin{aligned}\langle\psi|\psi\rangle &= (-i\langle\phi_1| - 3i\langle\phi_2| - \langle\phi_3|)(i|\phi_1\rangle + 3i|\phi_2\rangle - |\phi_3\rangle) \\ &= (-i)(i)\langle\phi_1|\phi_1\rangle + (-i)(3i)\langle\phi_1|\phi_2\rangle + (-i)(-1)\langle\phi_1|\phi_3\rangle \\ &\quad + (-3i)(i)\langle\phi_2|\phi_1\rangle + (-3i)(3i)\langle\phi_2|\phi_2\rangle + (-3i)(-1)\langle\phi_2|\phi_3\rangle \\ &\quad + (-1)(i)\langle\phi_3|\phi_1\rangle + (-1)(3i)\langle\phi_3|\phi_2\rangle + (-1)(-1)\langle\phi_3|\phi_3\rangle \\ &= 1 + 9 + 1 \\ &= 11,\end{aligned}$$

and

$$\begin{aligned}\langle\chi|\chi\rangle &= (\langle\phi_1| + i\langle\phi_2| - 5i\langle\phi_3|)(|\phi_1\rangle - i|\phi_2\rangle + 5i|\phi_3\rangle) \\ &= \langle\phi_1|\phi_1\rangle + \langle\phi_2|\phi_2\rangle + 25\langle\phi_3|\phi_3\rangle \\ &= 1 + 1 + 25 \\ &= 27.\end{aligned}$$

For a much faster method to calculate these norms, see 6.30 (b) in [Axl24]

Now, for $\langle\chi|\psi\rangle$ and $\langle\psi|\chi\rangle$:

$$\begin{aligned}
\langle \chi | \psi \rangle &= (\langle \phi_1 | + i\langle \phi_2 | - 5i\langle \phi_3 |)(i|\phi_1\rangle + 3i|\phi_2\rangle - |\phi_3\rangle) \\
&= i\langle \phi_1 | \phi_1 \rangle - 3\langle \phi_2 | \phi_2 \rangle + 5i\langle \phi_3 | \phi_3 \rangle \\
&= i - 3 + 5i \\
&= -3 + 6i,
\end{aligned}$$

and

$$\begin{aligned}
\langle \psi | \chi \rangle &= (-i\langle \phi_1 | - 3i\langle \phi_2 | - \langle \phi_3 |)(|\phi_1\rangle - i|\phi_2\rangle + 5i|\phi_3\rangle) \\
&= -i\langle \phi_1 | \phi_1 \rangle - 3\langle \phi_2 | \phi_2 \rangle - 5i\langle \phi_3 | \phi_3 \rangle \\
&= -i - 3 - 5i \\
&= -3 - 6i.
\end{aligned}$$

Note that the line of working for $\langle \psi | \chi \rangle$ is redundant because we could have taken the complex conjugate of $\langle \chi | \psi \rangle$.

As expected, we have that $\langle \chi | \psi \rangle = \langle \psi | \chi \rangle^*$.

For $\langle \psi + \chi | \psi + \chi \rangle$:

$$\begin{aligned}
\langle \psi + \chi | \psi + \chi \rangle &= \langle \psi | \psi \rangle + \langle \psi | \chi \rangle + \langle \chi | \psi \rangle + \langle \chi | \chi \rangle \\
&= 11 + (-3 - 6i) + (-3 + 6i) + 27 \\
&= 32.
\end{aligned}$$

Exercise 2.1(b)

To compute the operators $|\psi\rangle\langle\chi|$ and $|\chi\rangle\langle\psi|$:

$$\begin{aligned}
|\psi\rangle\langle\chi| &= (i|\phi_1\rangle + 3i|\phi_2\rangle - |\phi_3\rangle)(\langle\phi_1| + i\langle\phi_2| - 5i\langle\phi_3|) \\
&= i|\phi_1\rangle\langle\phi_1| + i^2|\phi_1\rangle\langle\phi_2| - 5i^2|\phi_1\rangle\langle\phi_3| \\
&\quad + 3i|\phi_2\rangle\langle\phi_1| + 3i^2|\phi_2\rangle\langle\phi_2| - 15i^2|\phi_2\rangle\langle\phi_3| \\
&\quad - |\phi_3\rangle\langle\phi_1| - i|\phi_3\rangle\langle\phi_2| + 5i|\phi_3\rangle\langle\phi_3| \\
&= i|\phi_1\rangle\langle\phi_1| - |\phi_1\rangle\langle\phi_2| + 5|\phi_1\rangle\langle\phi_3| \\
&\quad + 3i|\phi_2\rangle\langle\phi_1| - 3|\phi_2\rangle\langle\phi_2| + 15|\phi_2\rangle\langle\phi_3| \\
&\quad - |\phi_3\rangle\langle\phi_1| - i|\phi_3\rangle\langle\phi_2| + 5i|\phi_3\rangle\langle\phi_3|,
\end{aligned}$$

and

$$\begin{aligned}
|\chi\rangle\langle\psi| &= (|\phi_1\rangle - i|\phi_2\rangle + 5i|\phi_3\rangle)(-i\langle\phi_1| - 3i\langle\phi_2| - \langle\phi_3|) \\
&= -i|\phi_1\rangle\langle\phi_1| - 3i|\phi_1\rangle\langle\phi_2| - |\phi_1\rangle\langle\phi_3| \\
&\quad - i(-i)|\phi_2\rangle\langle\phi_1| - 3i(-i)|\phi_2\rangle\langle\phi_2| + i|\phi_2\rangle\langle\phi_3| \\
&\quad - 5i^2|\phi_3\rangle\langle\phi_1| - 15i^2|\phi_3\rangle\langle\phi_2| - 5i|\phi_3\rangle\langle\phi_3| \\
&= -i|\phi_1\rangle\langle\phi_1| - 3i|\phi_1\rangle\langle\phi_2| - |\phi_1\rangle\langle\phi_3| \\
&\quad - |\phi_2\rangle\langle\phi_1| - 3|\phi_2\rangle\langle\phi_2| + i|\phi_2\rangle\langle\phi_3| \\
&\quad + 5|\phi_3\rangle\langle\phi_1| + 15|\phi_3\rangle\langle\phi_2| - 5i|\phi_3\rangle\langle\phi_3|.
\end{aligned}$$

The traces are easy to compute, because $\text{Tr}(\hat{A}\hat{B}) = \text{Tr}(\hat{B}\hat{A})$ (for a proof, see 8.56 in [Axl24]):

$$\text{Tr}(|\psi\rangle\langle\chi|) = \text{Tr}(\langle\chi|\psi\rangle) = -3 + 6i$$

$$\text{Tr}(|\chi\rangle\langle\psi|) = \text{Tr}(\langle\psi|\chi\rangle) = -3 - 6i$$

Notice how the traces are complex conjugates of one another.

Exercise 2.1 (c)

The computation of the requested Hermitian conjugates of the quantum states are trivial.

Exercise 2.2

Consider the state vector $|\psi_1\rangle = |\phi_1\rangle + 4i|\phi_2\rangle + 5|\phi_3\rangle$. Taking complex conjugates and interchanging kets with bras, we obtain

$$\langle\psi_1| = \langle\phi_1| - 4i\langle\phi_2| + 5\langle\phi_3|,$$

Since $|\psi_2\rangle = b|\phi_1\rangle + 4|\phi_2\rangle - 3i|\phi_3\rangle$, we compute the inner product

$$\begin{aligned}
\langle\psi_1|\psi_2\rangle &= (\langle\phi_1| - 4i\langle\phi_2| + 5\langle\phi_3|)(b|\phi_1\rangle + 4|\phi_2\rangle - 3i|\phi_3\rangle) \\
&= b\langle\phi_1|\phi_1\rangle + 4\langle\phi_1|\phi_2\rangle - 3i\langle\phi_1|\phi_3\rangle \\
&\quad - 4ib\langle\phi_2|\phi_1\rangle - 16i\langle\phi_2|\phi_2\rangle + 12\langle\phi_2|\phi_3\rangle \\
&\quad + 5b\langle\phi_3|\phi_1\rangle + 20\langle\phi_3|\phi_2\rangle - 15i\langle\phi_3|\phi_3\rangle \\
&= b - 16i - 15i \\
&= b - 31i.
\end{aligned}$$

For the quantum states to be orthogonal, require that $\langle \psi_1 | \psi_2 \rangle := 0$, which forces $b = 31i$ by the equation above.

Again, for a much quicker calculation, see 6.30(c) in [Axl24].

Exercise 2.3

All the necessary products for you to verify (a) the Cauchy-Schwarz Inequality and (b) the Triangle Inequality are contained in Exercise 2.1(a).

These inequalities must always be true, by 6.14 and 6.17 in [Axl24], so the result is no surprise.

Exercise 2.4

Consider the quantum states

$$\begin{aligned} |\psi\rangle &= \alpha|\phi_1\rangle + 5|\phi_2\rangle \\ |\chi\rangle &= 3\alpha|\phi_1\rangle - 4|\phi_2\rangle. \end{aligned}$$

Taking the Hermitian conjugate of the state $|\psi\rangle$ yields the bra

$$\langle\psi| = \bar{\alpha}\langle\phi_1| + 5\langle\phi_2|,$$

so that

$$\begin{aligned} \langle\psi|\chi\rangle &= (\bar{\alpha}\langle\phi_1| + 5\langle\phi_2|)(3\alpha|\phi_1\rangle - 4|\phi_2\rangle) \\ &= 3\alpha\bar{\alpha}\langle\phi_1|\phi_1\rangle - 4\bar{\alpha}\langle\phi_1|\phi_2\rangle \\ &\quad + 15\alpha\langle\phi_2|\phi_1\rangle - 20\langle\phi_2|\phi_2\rangle \\ &= 3|\alpha|^2 - 20. \end{aligned}$$

In order for the states $|\psi\rangle$ and $|\chi\rangle$ to be orthogonal, mandate

$$\langle\psi|\chi\rangle = 0,$$

which forces

$$|\alpha|^2 = \frac{20}{3}.$$

Now for $\alpha \in \mathbb{R}$, we can either have $\alpha = \sqrt{\frac{20}{3}}$ or $\alpha = -\sqrt{\frac{20}{3}}$.

More generally, observe that, if $B(x, r)$ denotes the open ball of radius r around a point x in a metric space (X, d) , we see that for all

$$\alpha \in \partial B\left(\mathbf{0}_{\mathbb{C}}, \sqrt{20/3}\right),$$

the states $|\psi\rangle$ and $|\chi\rangle$ will be orthonormal.

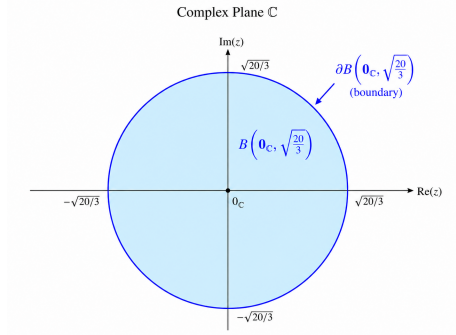


Figure 2.1: The complex ball centered about the origin (image generated by ChatGPT)

Exercise 2.5

Consider the quantum states $|\psi\rangle = |\phi_1\rangle + |\phi_2\rangle$ and $|\chi\rangle = |\phi_1\rangle - |\phi_2\rangle$, where $|\phi_1\rangle$ and $|\phi_2\rangle$ are *not* orthonormal. We claim that we have the following identities:

$$\begin{aligned}\langle\psi|\psi\rangle + \langle\chi|\chi\rangle &= 2\langle\phi_1|\phi_1\rangle + 2\langle\phi_2|\phi_2\rangle, \\ \langle\psi|\psi\rangle - \langle\chi|\chi\rangle &= 2\langle\phi_1|\phi_2\rangle + 2\langle\phi_2|\phi_1\rangle.\end{aligned}$$

Proof:

Taking the Hermitian conjugate of the state vector $|\psi\rangle = |\phi_1\rangle + |\phi_2\rangle$ yields the bra $\langle\psi| = \langle\phi_1| + \langle\phi_2|$. Consequently, the norm of $|\psi\rangle$ is given by

$$\begin{aligned}\langle\psi|\psi\rangle &= (\langle\phi_1| + \langle\phi_2|)(|\phi_1\rangle + |\phi_2\rangle) \\ &= \langle\phi_1|\phi_1\rangle + \langle\phi_1|\phi_2\rangle + \langle\phi_2|\phi_1\rangle + \langle\phi_2|\phi_2\rangle.\end{aligned}$$

Similarly, we have $\langle\chi| = \langle\phi_1| - \langle\phi_2|$,

which yields

$$\begin{aligned}\langle\chi|\chi\rangle &= (\langle\phi_1| - \langle\phi_2|)(|\phi_1\rangle - |\phi_2\rangle) \\ &= \langle\phi_1|\phi_1\rangle - \langle\phi_1|\phi_2\rangle - \langle\phi_2|\phi_1\rangle + \langle\phi_2|\phi_2\rangle.\end{aligned}$$

It immediately follows that

$$\begin{aligned}
\langle \psi | \psi \rangle + \langle \chi | \chi \rangle &= (\langle \phi_1 | \phi_1 \rangle + \langle \phi_1 | \phi_2 \rangle + \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&\quad + (\langle \phi_1 | \phi_1 \rangle - \langle \phi_1 | \phi_2 \rangle - \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&= 2\langle \phi_1 | \phi_1 \rangle + 2\langle \phi_2 | \phi_2 \rangle,
\end{aligned}$$

and

$$\begin{aligned}
\langle \psi | \psi \rangle - \langle \chi | \chi \rangle &= (\langle \phi_1 | \phi_1 \rangle + \langle \phi_1 | \phi_2 \rangle + \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&\quad - (\langle \phi_1 | \phi_1 \rangle - \langle \phi_1 | \phi_2 \rangle - \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&= 2\langle \phi_1 | \phi_2 \rangle + 2\langle \phi_2 | \phi_1 \rangle,
\end{aligned}$$

which is the desired result. ■

Exercise 2.6(a)

Consider the quantum state

$$|\psi\rangle = \frac{1}{\sqrt{15}}|\phi_1\rangle + \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle$$

where the kets $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$ are orthonormal.

By orthonormality, we have that

$$\begin{aligned}
\langle \psi | \psi \rangle &= \left(\frac{1}{\sqrt{15}} \right)^2 + \left(\frac{1}{\sqrt{3}} \right)^2 + \left(\frac{1}{\sqrt{5}} \right)^2 \\
&= \frac{1}{15} + \frac{1}{3} + \frac{1}{5} \\
&= \frac{9}{15} \\
&= \frac{3}{5}.
\end{aligned}$$

It follows the norm of $|\psi\rangle$ is given by

$$\|\psi\| = \sqrt{\frac{3}{5}}.$$

Exercise 2.6(b)

Given further that the $|\phi_n\rangle$ are the eigenstates of an operator $\hat{B}$, with $\hat{B}|\phi_n\rangle = (3n^2 - 1)|\phi_n\rangle$, for $n \in \{1, 2, 3\}$.

By linearity, one can verify that

$$\begin{aligned}
 \hat{B}|\psi\rangle &= \hat{B}\left(\frac{1}{\sqrt{15}}|\phi_1\rangle + \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle\right) \\
 &= \frac{1}{\sqrt{15}}\hat{B}|\phi_1\rangle + \frac{1}{\sqrt{3}}\hat{B}|\phi_2\rangle + \frac{1}{\sqrt{5}}\hat{B}|\phi_3\rangle \\
 &= \frac{2}{\sqrt{15}}|\phi_1\rangle + \frac{11}{\sqrt{3}}|\phi_2\rangle + \frac{26}{\sqrt{5}}|\phi_3\rangle.
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \langle\psi|\hat{B}|\psi\rangle &= \left(\frac{1}{\sqrt{15}}\langle\phi_1| + \frac{1}{\sqrt{3}}\langle\phi_2| + \frac{1}{\sqrt{5}}\langle\phi_3|\right)\left(\frac{2}{\sqrt{15}}|\phi_1\rangle + \frac{11}{\sqrt{3}}|\phi_2\rangle + \frac{26}{\sqrt{5}}|\phi_3\rangle\right) \\
 &= \frac{2}{15} + \frac{11}{3} + \frac{26}{5} \\
 &= \frac{2 + 55 + 78}{15} \\
 &= \frac{135}{15} \\
 &= 9.
 \end{aligned}$$

The expectation value of $\hat{B}$ with respect to the state $|\psi\rangle$ is then

$$\begin{aligned}
 \langle\hat{B}\rangle &= \frac{\langle\psi|\hat{B}|\psi\rangle}{\langle\psi|\psi\rangle} \\
 &= \frac{9}{3/5} \\
 &= 15.
 \end{aligned}$$

where the second equality is a consequence of our work in part (a).

Exercise 2.6(c)

$$\begin{aligned}
 \hat{B}^2|\psi\rangle &= \hat{B}^2\left(\frac{1}{\sqrt{15}}|\phi_1\rangle + \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle\right) \\
 &= \frac{1}{\sqrt{15}}\hat{B}^2|\phi_1\rangle + \frac{1}{\sqrt{3}}\hat{B}^2|\phi_2\rangle + \frac{1}{\sqrt{5}}\hat{B}^2|\phi_3\rangle \\
 &= \frac{2^2}{\sqrt{15}}|\phi_1\rangle + \frac{11^2}{\sqrt{3}}|\phi_2\rangle + \frac{26^2}{\sqrt{5}}|\phi_3\rangle, \quad (\text{See Theorem A.1.}) \\
 &= \frac{4}{\sqrt{15}}|\phi_1\rangle + \frac{121}{\sqrt{3}}|\phi_2\rangle + \frac{676}{\sqrt{5}}|\phi_3\rangle.
 \end{aligned}$$

$$\begin{aligned}
\langle \psi | \hat{B}^2 | \psi \rangle &= \left(\frac{1}{\sqrt{15}} \langle \phi_1 | + \frac{1}{\sqrt{3}} \langle \phi_2 | + \frac{1}{\sqrt{5}} \langle \phi_3 | \right) \left(\frac{4}{\sqrt{15}} | \phi_1 \rangle + \frac{121}{\sqrt{3}} | \phi_2 \rangle + \frac{676}{\sqrt{5}} | \phi_3 \rangle \right) \\
&= \frac{4}{15} + \frac{121}{3} + \frac{676}{5} \\
&= \frac{4 + 605 + 2028}{15} \\
&= \frac{2637}{15} \\
&= \frac{879}{5}.
\end{aligned}$$

It follows that the expectation value of $\langle \hat{B}^2 \rangle$ is given by

$$\begin{aligned}
\langle \hat{B}^2 \rangle &= \frac{\langle \psi | \hat{B}^2 | \psi \rangle}{\langle \psi | \psi \rangle} \\
&= \frac{\left(\frac{879}{5} \right)}{3/5} \\
&= 293.
\end{aligned}$$

Exercises 2.7, 2.8, 2.9 and 2.10

All trivial.

For Exercise 2.7(b), you may wish to use Euler's identity, and for Exercise 2.8(e), the hyperbolic tangent identities may be useful. See, for instance, the formulae in [Tay05]

Exercise 2.11

Theorem. If $\hat{A}$ is projection operator, then $1 - \hat{A}$ is a projection operator.

Proof: Define the operator $\hat{B} = \hat{I} - \hat{A}$. As $\hat{A}$ is a projection operator by assumption, and by the properties of the adjoint, we have

$$\hat{B}^\dagger = (\hat{I} - \hat{A})^\dagger = \hat{I}^\dagger - \hat{A}^\dagger = \hat{I} - \hat{A} := \hat{B},$$

and

$$\hat{B}^2 = (\hat{I} - \hat{A})^2 = \hat{I} - 2\hat{A}\hat{I} + \hat{A}^2 = \hat{I} - 2\hat{A} + \hat{A} = \hat{I} - \hat{A} := \hat{B}$$

so that, by definition, $\hat{B}$ constitutes a projection operator, as required. ■

Exercise 2.1.2.

Define the operator

$$\hat{A} := \frac{|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle}.$$

We propose the following result:

Theorem. $\hat{A}$ is a projection operator.

Proof: Indeed, it is clear that $\hat{A}$ is self-adjoint. Now, observe

$$\begin{aligned}\hat{A}^2 &= \left(\frac{|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle} \right) \left(\frac{|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle} \right)^1 \\ &= \frac{|\psi\rangle\langle\psi|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle^2} \\ &= \frac{|\psi\rangle\langle\psi|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle^2} \\ &= \frac{|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle} \\ &= \hat{A},\end{aligned}$$

which verifies our claim². ■

Exercise 2.13(a)

For $\langle\phi|\hat{A}|\psi\rangle\langle\psi|$,

$$\begin{aligned}\left(\langle\phi|\hat{A}|\psi\rangle\langle\psi|\right)^\dagger &= \langle\psi|^\dagger \left(\langle\phi|\hat{A}|\psi\rangle\right)^\dagger \\ &= |\psi\rangle\langle\psi|\hat{A}^\dagger|\phi\rangle.\end{aligned}$$

Exercise 2.13(b)

Consider now $\hat{A}|\psi\rangle\langle\phi|$. It is easy to see that this acts as an operator, as for any state $|\chi\rangle$,

$$\left(\hat{A}|\psi\rangle\langle\phi|\right)|\chi\rangle = (\langle\phi|\chi\rangle)\hat{A}|\psi\rangle.$$

¹When evaluating T^n where T is some operator, *always* write, for example, $T^2 = TT$.

²Notice that we never assumed that the ket $|\psi\rangle$ was normalised.

$$\begin{aligned} (\hat{A}|\psi\rangle\langle\phi|)^\dagger &= \langle\phi|^\dagger|\psi\rangle^\dagger\hat{A}^\dagger \\ &= |\phi\rangle\langle\psi|\hat{A}^\dagger. \end{aligned}$$

Exercise 2.13(c)

$$\begin{aligned} (\langle\phi|\hat{A}|\psi\rangle|\psi\rangle\langle\phi|\hat{A})^\dagger &= \hat{A}^\dagger(|\psi\rangle\langle\phi|)^\dagger(\langle\phi|\hat{A}|\psi\rangle)^\dagger \\ &= (\hat{A}^\dagger|\phi\rangle\langle\psi|)(\langle\psi|\hat{A}^\dagger|\phi\rangle). \end{aligned}$$

Exercise 2.13(d)

$$\begin{aligned} (\langle\psi|\hat{A}|\phi\rangle|\phi\rangle + i\hat{A}|\psi\rangle)^\dagger &= (\langle\psi|\hat{A}|\phi\rangle|\phi\rangle)^\dagger + (i\hat{A}|\psi\rangle)^\dagger \\ &= \langle\phi|(\langle\psi|\hat{A}|\phi\rangle)^\dagger - i\langle\psi|\hat{A}^\dagger \\ &= \langle\phi|\langle\phi|\hat{A}^\dagger|\psi\rangle - i\langle\psi|\hat{A}^\dagger. \end{aligned}$$

Exercise 2.13(e)

$$\begin{aligned} (|\phi\rangle\langle\phi|\hat{A} - i\hat{A}|\psi\rangle\langle\psi|)^\dagger &= (|\phi\rangle\langle\phi|\hat{A})^\dagger + (-i\hat{A}|\psi\rangle\langle\psi|)^\dagger \\ &= \hat{A}^\dagger|\phi\rangle\langle\phi| + i|\psi\rangle\langle\psi|\hat{A}^\dagger. \end{aligned}$$

For proofs of the properties of adjoints, see 7.5 in [Axl24].

Exercise 2.14(a)

Since $|\phi_1\rangle$ and $|\phi_2\rangle$ are orthonormal, and is the same length as the dimension of the vector space (by assumption), we have that the list $|\phi_1\rangle, |\phi_2\rangle$ forms a basis by 6.28 in [Axl24].

Exercise 2.14(b)

Consider the operator $\hat{B} = |\phi_1\rangle\langle\phi_2|$. We have

$$\hat{B}^\dagger = |\phi_2\rangle\langle\phi_1| \neq \hat{B},$$

so $\hat{B}$ is not Hermitian.

Now, observe that

$$\begin{aligned}\hat{B}^2 &= \hat{B}\hat{B} = (|\phi_1\rangle\langle\phi_2|)(|\phi_1\rangle\langle\phi_2|) \\ &= |\phi_1\rangle\langle\phi_2||\phi_1\rangle\langle\phi_2| = 0,\end{aligned}$$

by orthonormality.

Exercise 2.14(c)

Theorem. The products $\hat{B}\hat{B}^\dagger$ and $\hat{B}^\dagger\hat{B}$ are projection operators.

Proof: By definition,

$$(\hat{B}\hat{B}^\dagger)^\dagger = (\hat{B}^\dagger)^\dagger\hat{B}^\dagger = \hat{B}\hat{B}^\dagger,$$

and

$$(\hat{B}^\dagger\hat{B})^\dagger = \hat{B}^\dagger(\hat{B}^\dagger)^\dagger = \hat{B}^\dagger\hat{B}.$$

Hence, $\hat{B}\hat{B}^\dagger$ and $\hat{B}^\dagger\hat{B}$ are both Hermitian.

It is easy to verify from the definition of $\hat{B}$ that $\hat{B}\hat{B}^\dagger = |\phi_1\rangle\langle\phi_1|$ and $\hat{B}^\dagger\hat{B} = |\phi_2\rangle\langle\phi_2|$ (do so now). It immediately follows that $(\hat{B}\hat{B}^\dagger)^2 = \hat{B}\hat{B}^\dagger$, and $(\hat{B}^\dagger\hat{B})^2 = \hat{B}^\dagger\hat{B}$ (why?) so that the two products constitute projection operators, as required. ■

Exercise 2.14(d)

Theorem. The operator $\hat{B}\hat{B}^\dagger - \hat{B}^\dagger\hat{B}$ is unitary.

Proof: Define $\hat{T} = \hat{B}\hat{B}^\dagger - \hat{B}^\dagger\hat{B}$. Our work above shows that

$$\hat{T} = |\phi_1\rangle\langle\phi_1| - |\phi_2\rangle\langle\phi_2|.$$

It follows that $\hat{T}|\phi_1\rangle = |\phi_1\rangle$ and $\hat{T}|\phi_2\rangle = -|\phi_2\rangle$. The matrix of $\hat{T}$ with respect to the orthonormal basis $|\phi_1\rangle, |\phi_2\rangle$ is, therefore

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

From 7.53 of [Axl24], we see that $\hat{T}$ is unitary, as required. ■

Exercise 2.14(e)

Trivial.

Exercise 2.16(a)

Consider the matrix

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

Its columns are given by $a_1 = (\cos \theta, \sin \theta)$ and $a_2 = (-\sin \theta, \cos \theta)$. With respect to the standard Euclidean norm, the Pythagorean identity shows that

$$\|a_1\| = \|a_2\| = 1,$$

and it is also easy to deduce that $\langle a_1, a_2 \rangle = 0$, i.e. the columns form an orthonormal list in $\mathbb{R}^2$, so A is a unitary matrix by 7.57(b) of [Ax124].

Exercise 2.16(b)

We treat the case $\theta \neq 0$, otherwise A is simply the identity.

To find the eigenvalues of A amounts to solving the equation $\det(A - \lambda I) = 0$ for λ . This is an easy computation, and we obtain

$$\lambda = e^{\pm i\theta},$$

to which we can find a pair of normalised eigenvectors

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \text{ and } v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix},$$

corresponding to the eigenvalues $\lambda_+ = e^{i\theta}$ and $\lambda_- = e^{-i\theta}$, respectively.

Remark: You might notice some similarity between A and the ordinary rotational matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The difference is that the matrix above represents a *counterclockwise* rotation about the origin, whereas A represents a *clockwise* rotation about the origin (see the figure below). That A has no real eigenvalues, except when $\theta = 0$, aligns with our intuition: Multiplication of a vector by a real scalar will not cause it to rotate!

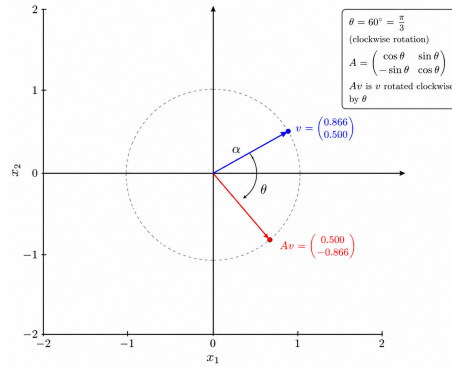


Figure 2.2: The action of A on a vector about the origin (image generated by ChatGPT)

Exercises 2.17, 2.18 and 2.19

All are computationally heavy, and you should be able to figure these out yourself by referring to the book when necessary.

Exercise 2.20

We claim to possess the following result:

Let $\hat{A}$, $\hat{B}$, and $\hat{C}$ be operators. Then we have

$$\text{Tr}(\hat{A}\hat{B}\hat{C}) = \text{Tr}(\hat{C}\hat{A}\hat{B}) = \text{Tr}(\hat{B}\hat{C}\hat{A}).$$

Proof: Note the identity $\text{Tr}(\hat{S}\hat{T}) = \text{Tr}(\hat{T}\hat{S})$. Observe that

$$\begin{aligned} \text{Tr}(\hat{A}\hat{B}\hat{C}) &= \text{Tr}((\hat{A}\hat{B})\hat{C}) = \text{Tr}(\hat{C}(\hat{A}\hat{B})) = \text{Tr}(\hat{C}\hat{A}\hat{B}) \\ &= \text{Tr}((\hat{C}\hat{A})\hat{B}) = \text{Tr}(\hat{B}(\hat{C}\hat{A})) = \text{Tr}(\hat{B}\hat{C}\hat{A}), \end{aligned}$$

which completes the proof. ■

Exercises 2.21, 2.22, 2.23, 2.24

All are computationally heavy, and you should be able to figure these out by referring to the book when necessary³.

³Unless you are a masochist, I would recommend using software to calculate the eigenvalues for you.

Exercise 2.25 Consider an operator $\hat{A}$ so that

$$[\hat{A}, \hat{A}^\dagger] = 1.$$

(a) Evaluate the commutators

$$[\hat{A}^\dagger \hat{A}, \hat{A}] \quad \text{and} \quad [\hat{A}^\dagger \hat{A}, \hat{A}^\dagger].$$

(b) If the actions of $\hat{A}$ and $\hat{A}^\dagger$ on the states $\{|a\rangle\}$ are given by

$$\hat{A}|a\rangle = \sqrt{a}|a-1\rangle$$

and

$$\hat{A}^\dagger|a\rangle = \sqrt{a+1}|a+1\rangle$$

and if $\langle a'|a\rangle = \delta_{a'a}$, calculate

$$\langle a|\hat{A}|a+1\rangle, \quad \langle a+1|\hat{A}^\dagger|a\rangle$$

and

$$\langle a|\hat{A}^\dagger \hat{A}|a\rangle, \quad \langle a|\hat{A} \hat{A}^\dagger|a\rangle.$$

(c) Calculate

$$\langle a|(\hat{A} + \hat{A}^\dagger)^2|a\rangle \quad \text{and} \quad \langle a|(\hat{A} - \hat{A}^\dagger)^2|a\rangle.$$

(a) Using the distributive properties of the commutator:

$$[\hat{A}^\dagger \hat{A}, \hat{A}] = \hat{A}^\dagger [\hat{A}, \hat{A}] + [\hat{A}^\dagger, \hat{A}] \hat{A} = -\hat{A}$$

where the last equality follows from anticommutativity.

For the second one:

$$[\hat{A}^\dagger \hat{A}, \hat{A}^\dagger] = \hat{A}^\dagger [\hat{A}, \hat{A}^\dagger] + [\hat{A}^\dagger, \hat{A}^\dagger] \hat{A} = \hat{A}^\dagger.$$

(b)

These are trivial calculations.

By definition,

$$\begin{aligned} \langle a|\hat{A}|a+1\rangle &= \langle a|\sqrt{a+1}|a\rangle \\ &= \sqrt{a+1}\langle a|a\rangle \\ &= \sqrt{a+1}. \end{aligned}$$

Similarly,

$$\begin{aligned}\langle a+1|\hat{A}^\dagger|a\rangle &= \langle a+1|\sqrt{a+1}|a+1\rangle \\ &= \sqrt{a+1}\langle a+1|a+1\rangle \\ &= \sqrt{a+1}.\end{aligned}$$

For the first product,

$$\begin{aligned}\langle a|\hat{A}^\dagger\hat{A}|a\rangle &= \langle a|\hat{A}^\dagger(\sqrt{a}|a-1\rangle) \\ &= \sqrt{a}\langle a|\hat{A}^\dagger|a-1\rangle \\ &= \sqrt{a}\langle a|\sqrt{a}|a\rangle \\ &= a\langle a|a\rangle \\ &= a.\end{aligned}$$

For the second product,

$$\begin{aligned}\langle a|\hat{A}\hat{A}^\dagger|a\rangle &= \langle a|\hat{A}(\sqrt{a+1}|a+1\rangle) \\ &= \sqrt{a+1}\langle a|\hat{A}|a+1\rangle \\ &= a+1.\end{aligned}$$

*Recall that the Kronecker Delta is defined as $\delta_{a,a'} = \begin{cases} 1, & a = a' \\ 0, & a \neq a'. \end{cases}$

(c)

Before directly attacking the problem, we may wish to evaluate

$$(\hat{A} + \hat{A}^\dagger)^2 = (\hat{A} + \hat{A}^\dagger)(\hat{A} + \hat{A}^\dagger) = \hat{A}^2 + \hat{A}\hat{A}^\dagger + \hat{A}^\dagger\hat{A} + (\hat{A}^\dagger)^2$$

4

$$(\hat{A} - \hat{A}^\dagger)^2 = (\hat{A} - \hat{A}^\dagger)(\hat{A} - \hat{A}^\dagger) = \hat{A}^2 - \hat{A}\hat{A}^\dagger - \hat{A}^\dagger\hat{A} + (\hat{A}^\dagger)^2$$

$$\begin{aligned}\langle a|(\hat{A} + \hat{A}^\dagger)^2|a\rangle &= \langle a|\left(\hat{A}^2 + \hat{A}\hat{A}^\dagger + \hat{A}^\dagger\hat{A} + (\hat{A}^\dagger)^2\right)|a\rangle \\ &= \langle a|\hat{A}^2|a\rangle + \langle a|\hat{A}\hat{A}^\dagger|a\rangle + \langle a|\hat{A}^\dagger\hat{A}|a\rangle + \langle a|(\hat{A}^\dagger)^2|a\rangle \\ &= 0 + (a+1) + a + 0 \\ &= 2a+1.\end{aligned}$$

⁴Why can't I combine $\hat{A}\hat{A}^\dagger + \hat{A}^\dagger\hat{A}$? Under what conditions can I do so?

I want you to think about why the first and last terms go to zero.

Hint: what are the properties of the Kronecker Delta?

Similarly,

$$\begin{aligned}
 \langle a | (\hat{A} - \hat{A}^\dagger)^2 | a \rangle &= \langle a | \left(\hat{A}^2 - \hat{A}\hat{A}^\dagger - \hat{A}^\dagger\hat{A} + (\hat{A}^\dagger)^2 \right) | a \rangle \\
 &= \langle a | \hat{A}^2 | a \rangle - \langle a | \hat{A}\hat{A}^\dagger | a \rangle - \langle a | \hat{A}^\dagger\hat{A} | a \rangle + \langle a | (\hat{A}^\dagger)^2 | a \rangle \\
 &= 0 - (a + 1) - a + 0 \\
 &= -2a - 1.
 \end{aligned}$$

Exercise 2.28 Let A and B be two $n \times n$ matrices. Assuming that B^{-1} exists, show that

$$[A, B^{-1}] = -B^{-1}[A, B]B^{-1}.$$

Proof:

$$\begin{aligned}
 -B^{-1}[A, B]B^{-1} &= -B^{-1}(AB - BA)B^{-1} = -B^{-1}(A - BAB^{-1}) \\
 &= AB^{-1} - B^{-1}A := [A, B^{-1}].
 \end{aligned}$$

Exercise 2.33 Show that

$$[\hat{X}, \hat{P}^n] = i\hbar n \hat{P}^{n-1}.$$

Proof:

We have the relation $[\hat{X}, \hat{P}] = i\hbar$. Therefore,

$$\begin{aligned}
 [\hat{X}, \hat{P}^n] &= \sum_{j=0}^{n-1} \hat{P}^j [\hat{X}, \hat{P}] \hat{P}^{n-j-1} = i\hbar \left(\sum_{j=0}^{n-1} \hat{P}^{n-1} \right) \\
 &= i\hbar \left(\sum_{j=0}^{n-1} 1 \right) \hat{P}^{n-1} \\
 &= i\hbar n \hat{P}^{n-1}
 \end{aligned}$$

which completes the proof. ■

Appendix A

For the following result, I use the same notation as [Axl24].

Theorem A.1. Suppose $T \in \mathcal{L}(V)$, and v is an eigenvector of T corresponding to the eigenvalue λ , that is, $v \in V$ is such that $Tv = \lambda v$. Then for $n \in \mathbb{Z}^+$, we have $T^n v = \lambda^n v$.

Proof: The proof is by induction. Clearly the result is true for $n = 1$, so for our inductive step, suppose that this holds for some $n = k \geq 1$. Then, for $n = k + 1$, we have, by linearity,

$$T^n v = T(T^k v) = T(\lambda^k v) = \lambda^k T v = \lambda^k (\lambda v) = \lambda^{k+1} v = \lambda^n v,$$

which completes the proof. ■

References

- [Axl24] Sheldon Axler. *Linear Algebra Done Right*. Undergraduate Texts in Mathematics. Springer, 2024. ISBN: 978-3031410253.
- [SS03] Stein and Shakarchi. *Fourier Analysis*. Princeton University Press, 2003. ISBN: 978-0-691-11384-5.
- [Tay05] John R. Taylor. *Classical Mechanics*. University Science Books, 2005. ISBN: 978-1891389221.
- [Zet22] Nouredine Zettili. *Quantum Mechanics*. Wiley, 2022. ISBN: 978-1-118-30789-2.